NAME: ____________________ PERIOD: ________ DATE: ____________

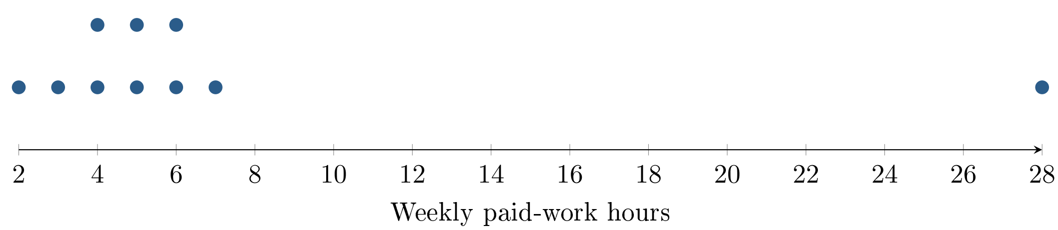
Typical Hours, Exceptional Week

AP Statistics · Unit 1 · Topics 1.7–1.8 (CED 1.7–1.8) · E: 2026-09-23

[STAR] TODAY'S PROBLEM

Suppose ten students report weekly paid-work hours: 2, 3, 4, 4, 5, 5, 6, 6, 7, 28. The mean is 7 hours.

Zoe: "Use the median and IQR with a boxplot; the boxplot will keep every individual hour value. " **Malik:** "Report the mean: a typical student works 7 hours, because the mean uses every observation."



FIRST TAKE — before we discuss (5 min)

Which report would you send to the principal, or would you revise both? Cite a feature of the data.

[BOOK] TWO TOPICS, ONE DATA STORY

1.7: Median and IQR resist extremes; mean and SD are more sensitive. Here quartiles are the medians of the two halves.
1.8: A modified boxplot marks values beyond $Q_1 - 1.5\text{IQR}$ or $Q_3 + 1.5\text{IQR}$ separately.
Whiskers reach the most extreme nonoutliers; the box spans the middle half and can hide repeats and gaps.

Work (a)–(c) from the rules box:

DOK 1 **(a)** State the median weekly work hours.

DOK 2 **(b)** Use medians of the lower and upper five values to find Q_1, Q_3, and IQR. Apply the $1.5\,\text{IQR}$ rule and sketch a modified boxplot, labeling the whiskers and any outlier.

DOK 3 **(c)** $\star$ In three to five sentences, evaluate both reports and propose a corrected report of typical hours and spread. Use the unusual value to explain your summary choice. Decide whether to keep the boxplot or dotplot, stating one benefit and one information limit of your choice.

Frame: I would report ____________________ because ____________________________________; the display cannot establish ___________________________.

EXIT — turn this in

One claim I would revise after reading the rules box: ____